

Perception of Time in Virtual Reality with Different Sensory Stimulation

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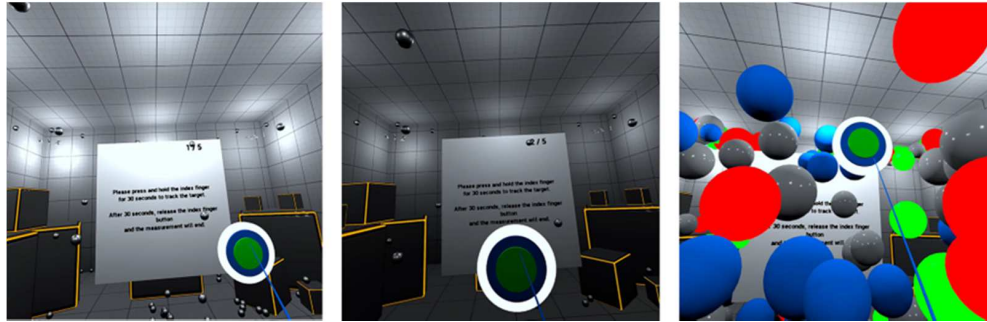


Figure 1: Participants were observed for a period of 30 seconds while they were exposed to a variety of sensory stimuli within a virtual environment. Left: Time measurement without stimulus conditions; Middle: Time measurement with visual sensory stimuli; Right: Time measurement in terms of the occurrence of events.

ABSTRACT

This paper investigates the phenomenon of perceptual distortion of time caused by the stimulation of the senses in a virtual environment. We examine how time perception changes with sensory stimuli at varying intervals, focusing on visual, auditory, tactile, and visual-auditory stimuli, as well as the impact of complex visual event frequencies. Participants' time perception before and during the experiment was compared. The analysis showed that visual stimuli induced time compression, and auditory stimuli induced time dilation. Time dilation was more effectively induced than time compression. Based on these results, personalizing the time interval of sensory stimuli may maximize the time distortion effect.

Index terms: Virtual Reality, Time Perception.

1 INTRODUCTION

Virtual reality(VR) provides the user with a multisensory experience, which creates the illusion of being in another space. Along with spatial illusion, it is possible to give temporal illusion in VR. For example, differences in the speed at which the virtual sun moves serve as zeitgebers, influencing human time perception[1]. Time perception in VR can be changed by external stimuli.

Different temporal experiences could be given from the interval between sensory stimuli, often influenced by multimodal stimulation[2]. By stimulating multiple senses simultaneously, multimodal stimulation can change the user's perception of time. Several studies have demonstrated that the interval and frequency of external sensory stimuli affect people's perception of time.

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Nevertheless, there is a lack of information regarding which sensory stimuli are effective in time illusion and to what extent the differences in individual time perception abilities affect it. Therefore, we conducted an experiment to investigate the time compression or time dilation effect by the type and interval of external sensory stimulation.

In this study, sensory stimuli are classified into five categories: visual, auditory, tactile, visual-auditory, and event occurrence. Sensory stimuli are applied at different intervals in VR, with the aim of inducing time compression and time dilation. Results are compared with the differences according to the type of sensory stimulation and the length of the interval. Experiments illustrate an effective sensory stimulation method for changing time perception in a virtual environment.

2 METHODOLOGY

2.1 Hypotheses and Participants

In this study, we examine the effect of visual, visual-auditory, auditory, and tactile stimuli and the occurrence of events on time perception. In addition, the study investigates the ways in which these effects apply to the varying time perception abilities of different people. In the hypothesis, “slow stimuli” refer to long intervals and low event occurrence, and “fast stimuli” refer to short intervals and high event occurrence. Our hypotheses are as follows:

(H1) Multiple modalities presented simultaneously have a greater effect on changes in time perception than a single modality presented individually.

(H1') The effects of visual, auditory, and tactile senses and the occurrence of events on time perception is different for each.

(H2) Fast stimuli induces time compression and is more effective in distorting time perception than slow stimuli that induces time dilation.

(H3) People who usually perceive time quickly or slowly will show similar effects of time perception in a virtual environment.

The participants were 31, and were mainly in their 20s (58.06%), followed by those in their 30s (22.58%), 50s (12.09%), and 40s (6.45%). It should be noted that precise age data was not collated, and instead participants were surveyed in broad age groups in order to identify general demographic trends. This research did not require Ethical Review Board (IRB) approval as it did not involve invasive human experiments.

2.2 Experimental Design

The virtual reality experimental environment was created and provided with the head-mounted display (Meta Quest 3). A stationary virtual avatar was used which has virtual hands only. Given the possibility that participants may become aware of the sensory stimulation if the experiment is repeated, only three randomly selected sensory conditions out of ten were conducted. To prevent unintended time dilation from boredom, 30 seconds was set as the reference time, and participants were tasked with aiming at a moving target to discourage them from counting seconds. The experimental flow is as follows: at first, the participants predict and record 30 seconds. This 30 seconds is a subjective 30 seconds. In the subsequent virtual room, they predict 30 seconds under each of the three random sensory conditions, and then check the time measured so far.

In order to verify H1, single modalities included visual, auditory, tactile senses, and event occurrence, while multi-modalities included visual-auditory senses. In order to verify H2, stimuli were categorized as 'fast' (950ms intervals) and 'slow' (1050ms intervals). The visual condition was implemented as a flash of light, the auditory condition as a buzzer sound, and the tactile condition as a controller vibration. The occurrence of events was implemented as the frequency, size, and color changes of balls being fired from the ceiling. To verify H3, the participants were divided into those with fast, slow, and natural time based on the subjective 30 seconds measured first. The effects of fast and slow stimuli, as measured by the difference in time perception, were then compared through the results of measuring 30 seconds under the sensory conditions and the questionnaire.

3 RESULTS

The 'time distortion percentage' is defined as the percentage of time measured by each participant in the sensory condition, divided by the initial 30 seconds. This term is used to describe the extent to which sensory stimuli affect time perception, as illustrated in Figure 2, which shows a dense distribution of the time distortion percentage for fast visual and slow auditory stimuli. In contrast, for visual-auditory stimuli, the distortion percentage was less than 100%, suggesting that time compression occurred even in slow stimuli, which contradicts H1. A thorough examination of the median, mean, and range of the distortion percentage values indicates that vision is the most effective sense for time compression and hearing is the most effective sense for time dilation. This partially supports H1' for the visual, auditory, tactile, and visual-auditory senses. As demonstrated in Figure 3, slow stimuli exhibited relatively distinct time distortion, while fast stimuli demonstrated a wide range of effects, with each individual experiencing time compression to different degrees. These findings suggest that slow stimuli are universally effective for time distortion, which does not support H2.

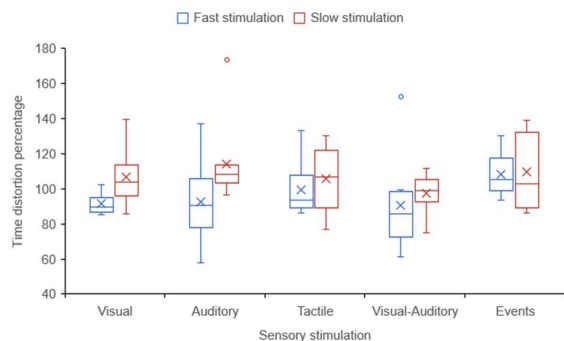


Figure 2: Time distortion percentage by sensory stimulation

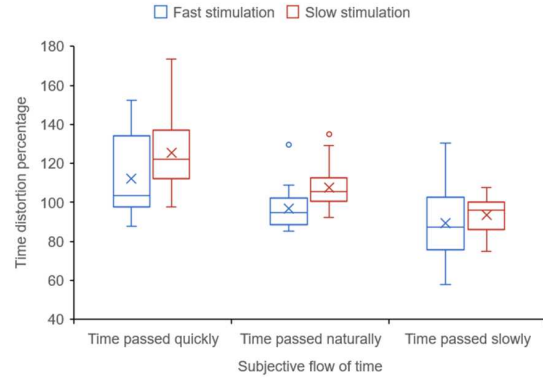


Figure 3: Time distortion percentage by subjective flow of time

As demonstrated in Figure 3, the 'Time passed naturally' group exhibited a 5% time compression or dilation, consistent with the experiment's objective. In contrast, the 'Time passed quickly' group demonstrated time dilation, attributed to their perception of all sensory stimuli as slow. Conversely, the 'Time passed slowly' group exhibited time compression, attributable to their perception of sensory stimuli as rapid. This finding supports H3, suggesting that individuals' time perception is influenced even in virtual environments.

4 CONCLUSION

The study demonstrated that by stimulating the senses at regular time intervals, time compression and time dilation effects can be induced, thereby distorting human time perception. Visual stimulation was effective in inducing time compression, and auditory stimulation was effective in inducing time dilation.

The participants' time perception was distinguished by the subjective 30 seconds measured at the beginning, and the degree of time distortion was found to vary depending on each person's time perception. In order to affect time perception, it is necessary to reduce the stimulus interval for those who pass time quickly, and to increase it for those who pass time slowly. Since time perception varies from person to person, setting a stimulus time interval that matches each individual's time perception characteristics can maximize the time distortion effect. This can contribute to improving time-sensitive interactions by collecting bias in the user's sense of time in virtual reality content, for example. The process for personalization requires further research.

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